Rheumatoid arthritis (RA) is an autoimmune disease that causes chronic synovial (joint) inflammation. RA immunopathogenesis involves LFA-1, a surface receptor found on leukocytes that is composed of an α-chain (CD11a) and a β-chain (CD18). The binding of LFA-1 to ICAM-1, an adhesion protein on the surface of B cells and endothelial cells, stimulates the migration of leukocytes to specific areas of inflammation. After successful LFA-1/ICAM-1 binding, CD18 is released into the plasma and synovial fluid (SF), where it exists in soluble form (sCD18). The levels of sCD18 in SF can be used as a marker for inflammation in RA.

CD18 release has been shown to increase in response to tumor necrosis factor (TNF), a pro-inflammatory cytokine involved in the inflammatory response of RA. In patients with newly diagnosed and untreated RA (eRA), leukocytes produce adequate quantities of LFA-1 and plasma sCD18 levels are unchanged (ie, elevated). However, patients with chronic RA (cRA) experience dysregulation of LFA-1 synthesis and an increase in competition for ICAM-1 by other ligands. Over time, this leads to a reduction in CD18 shedding and, as a result, a decrease in sCD18.

Experiment 1

An immunofluorescence assay was used to measure plasma sCD18 levels in patients with eRA and cRA, and in age- and gender-matched healthy controls (HC).

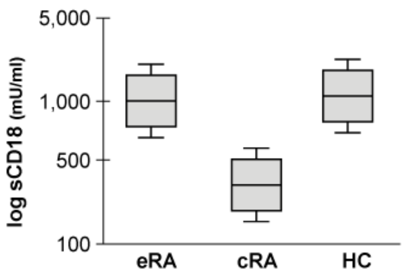


Figure 1 Baseline sCD18 measurements (Note: 50% of data points are below the median value; median = line inside the box.)

Experiment 2

Clinical assessments were performed in eRA patients to establish a baseline disease status. eRA patients were assigned a disease activity score based on characteristics of 28 joints and levels of C-reactive protein (DAS28CRP). This was followed by treatment with anti-inflammatory drugs. DAS28CRP scores were calculated at 3-month intervals for 12 months, with lower scores indicating decreased RA severity (Figure 2).

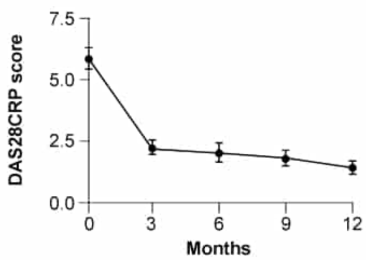


Figure 2 DAS28CRP scores in eRA patients

Experiment 3

Heterogeneous populations of mononuclear cells were cultured from the synovial fluid (SFMC) and peripheral blood (PBMC) of eRA patients during an acute exacerbation of their symptoms. sCD18 levels were measured in SFMCs and PBMCs that were untreated, treated with TNF, or treated with monoclonal antibodies targeting TNF (anti-TNF).

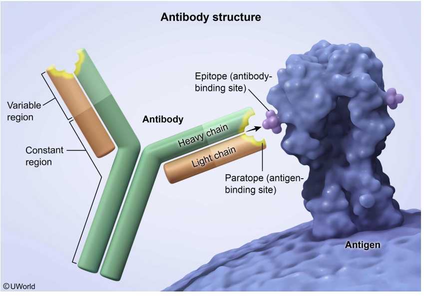
The antibodies used in Experiment 3 are composed of:

- A. two heavy chains that differ in amino acid sequence. [2%]
- B. two light chains that differ in amino acid sequence. [4%]
- ✔ C. a variable region that binds only one type of antigen. [74%]
- D. a constant region that binds only one type of antigen. [18%]

Correct

73% answered correctly

Explanation:



Antigens are foreign substances capable of triggering an adaptive immune response through antibody production. **Antibodies** are Y-shaped proteins that bind and neutralize antigens in the blood and lymph, preventing their entry into host cells and marking them for destruction (eg, phagocytosis/opsonization). Antibodies are produced and secreted by effector B cells (plasma cells), which are part of the adaptive immune system.

The **variable region** of the antibody binds the antigen. A single antibody *can bind only one type* of antigen, and this specificity is determined by the unique amino acid sequence of the antibody's variable region. Accordingly, antibodies that recognize different antigens have different variable regions. In this case, the **variable region** of the monoclonal antibody used in Experiment 3 is designed to specifically **bind only TNF** (ie, one type of antigen).

(Choices A and B) Most antibodies are made up of four polypeptide chains held together by disulfide bonds and noncovalent interactions. Specifically, antibodies consist of two identical heavy polypeptide chains and two identical light polypeptide chains. Together, the light and heavy chains have a constant region at one end and a variable region at the other end. Therefore, the antibodies used in Experiment 3 are composed of two heavy chains that would have identical (*not* different) amino acid sequences. Likewise, the two light chains of the antibodies would have identical (*not* different) amino acid sequences.

(Choice D) The constant region of antibodies *does not bind antigens*. Instead, the constant region determines antibody class and function by dictating which body cells and proteins the antibody can interact with to facilitate antigen destruction. The amino acid sequence of the constant region is largely conserved across all antibodies of the same class. Therefore, the antibodies used in Experiment 3 are composed of a constant region that does *not* bind antigens.

Educational objective:
Antibodies are synthesized and secreted by effector B cells, which are part of the adaptive immune system. Each antibody consists of two identical light chains, two identical heavy chains, a variable region that interacts with a specific antigen, and a constant region that interacts with the cells (eg, phagocytes) and proteins of the body to facilitate antigen destruction.

Time spent:0 sec
Subject: Biology

QID:400807
System: Skin and Immune Systems

Rheumatoid arthritis (RA) is an autoimmune disease that causes chronic synovial (joint) inflammation. RA immunopathogenesis involves LFA-1, a surface receptor found on leukocytes that is composed of an α-chain (CD11a) and a β-chain (CD18). The binding of LFA-1 to ICAM-1, an adhesion protein on the surface of B cells and endothelial cells, stimulates the migration of leukocytes to specific areas of inflammation. After successful LFA-1/ICAM-1 binding, CD18 is released into the plasma and synovial fluid (SF), where it exists in soluble form (sCD18). The levels of sCD18 in SF can be used as a marker for inflammation in RA.

CD18 release has been shown to increase in response to tumor necrosis factor (TNF), a pro-inflammatory cytokine involved in the inflammatory response of RA. In patients with newly diagnosed and untreated RA (eRA), leukocytes produce adequate quantities of LFA-1 and plasma sCD18 levels are unchanged (ie, elevated). However, patients with chronic RA (cRA) experience dysregulation of LFA-1 synthesis and an increase in competition for ICAM-1 by other ligands. Over time, this leads to a reduction in CD18 shedding and, as a result, a decrease in sCD18.

Experiment 1

An immunofluorescence assay was used to measure plasma sCD18 levels in patients with eRA and cRA, and in age- and gender-matched healthy controls (HC).

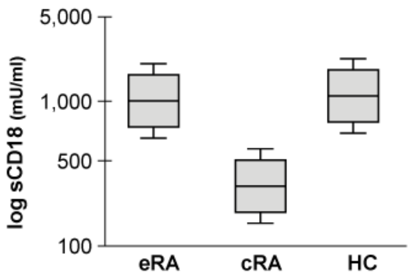


Figure 1 Baseline sCD18 measurements (Note: 50% of data points are below the median value; median = line inside the box.)

Experiment 2

Clinical assessments were performed in eRA patients to establish a baseline disease status. eRA patients were assigned a disease activity score based on characteristics of 28 joints and levels of C-reactive protein (DAS28CRP). This was followed by treatment with anti-inflammatory drugs. DAS28CRP scores were calculated at 3-month intervals for 12 months, with lower scores indicating decreased RA severity (Figure 2).

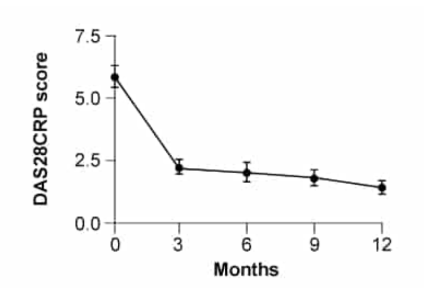


Figure 2 DAS28CRP scores in eRA patients

Experiment 3

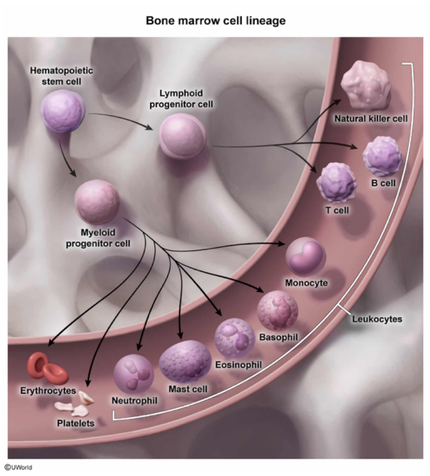
Heterogeneous populations of mononuclear cells were cultured from the synovial fluid (SFMC) and peripheral blood (PBMC) of eRA patients during an acute exacerbation of their symptoms. sCD18 levels were measured in SFMCs and PBMCs that were untreated, treated with TNF, or treated with monoclonal antibodies targeting TNF (anti-TNF).

According to the passage, cells containing the LFA-1 complex on their cell membranes most likely originate in which of the following tissues?

- A. Spleen [5%]
- B. Thymus [7%]
- ✔ C. Bone marrow [76%]
- D. Lymph nodes [11%]

Correct
76% answered correctly

Explanation:



The **bone marrow** is a soft, spongy tissue that lines the inside of bones and gives rise to most cells involved in the immune response. Cells originating in the bone marrow begin as multipotential hematopoietic stem cells and can differentiate into lymphoid or myeloid progenitor cells.

- **Lymphoid progenitor cells** go on to become B cells, T cells, or natural killer cells.
- **Myeloid progenitor cells** in the bone marrow further differentiate into erythrocytes, platelet-producing megakaryocytes, neutrophils, basophils, eosinophils, monocytes, or mast cells.

The passage states that LFA-1 is located on the surface of leukocytes (ie, white blood cells). Leukocytes include

lymphocytes, neutrophils, eosinophils, basophils, and monocytes. All leukocytes originate in the bone marrow. Therefore, cells **expressing LFA-1** on their surface **originate** from the **bone marrow** and *not* from the spleen, thymus, or lymph nodes (ie, other components of the lymphatic system).

(Choice A) The spleen is an abdominal organ that functions to filter blood, store erythrocytes, and destroy old erythrocytes. Active B cells can also be found in the spleen.

(Choice B) The thymus is a lymphoid organ that serves as the site for T cell maturation. T cells that contain LFA-1 originate in the bone marrow and later migrate to the thymus to mature.

(Choice D) Lymph nodes are small nodular lymphoid tissues ubiquitous in the human body. Lymph nodes monitor lymph (lymphatic fluid) for antigens and activate lymphocytes upon antigen detection. Leukocytes found in lymph nodes migrate there after maturation in other lymphoid organs.

Educational objective:
Hematopoietic stem cells originate in the bone marrow and differentiate into myeloid or lymphoid progenitor cells. Lymphoid progenitor cells go on to become T cells, B cells, or natural killer cells whereas myeloid progenitor cells differentiate into erythrocytes, megakaryocytes, neutrophils, basophils, eosinophils, monocytes, or mast cells.

Time spent:0 sec
Subject:
Biology

QID:400808
System:
Skin and Immune Systems

Rheumatoid arthritis (RA) is an autoimmune disease that causes chronic synovial (joint) inflammation. RA immunopathogenesis involves LFA-1, a surface receptor found on leukocytes that is composed of an α-chain (CD11a) and a β-chain (CD18). The binding of LFA-1 to ICAM-1, an adhesion protein on the surface of B cells and endothelial cells, stimulates the migration of leukocytes to specific areas of inflammation. After successful LFA-1/ICAM-1 binding, CD18 is released into the plasma and synovial fluid (SF), where it exists in soluble form (sCD18). The levels of sCD18 in SF can be used as a marker for inflammation in RA.

CD18 release has been shown to increase in response to tumor necrosis factor (TNF), a pro-inflammatory cytokine involved in the inflammatory response of RA. In patients with newly diagnosed and untreated RA (eRA), leukocytes produce adequate quantities of LFA-1 and plasma sCD18 levels are unchanged (ie, elevated). However, patients with chronic RA (cRA) experience dysregulation of LFA-1 synthesis and an increase in competition for ICAM-1 by other ligands. Over time, this leads to a reduction in CD18 shedding and, as a result, a decrease in sCD18.

Experiment 1

An immunofluorescence assay was used to measure plasma sCD18 levels in patients with eRA and cRA, and in age- and gender-matched healthy controls (HC).

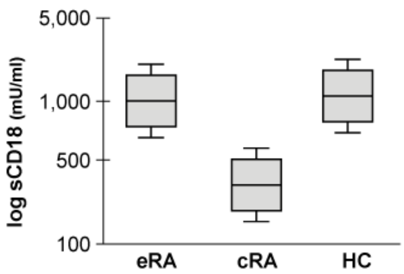


Figure 1 Baseline sCD18 measurements (Note: 50% of data points are below the median value; median = line inside the box.)

Experiment 2

Clinical assessments were performed in eRA patients to establish a baseline disease status. eRA patients were assigned a disease activity score based on characteristics of 28 joints and levels of C-reactive protein (DAS28CRP). This was followed by treatment with anti-inflammatory drugs. DAS28CRP scores were calculated at 3-month intervals for 12 months, with lower scores indicating decreased RA severity (Figure 2).

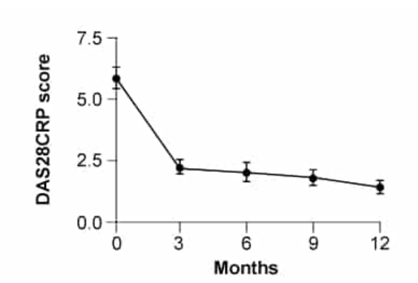


Figure 2 DAS28CRP scores in eRA patients

Experiment 3

Heterogeneous populations of mononuclear cells were cultured from the synovial fluid (SFMC) and peripheral blood (PBMC) of eRA patients during an acute exacerbation of their symptoms. sCD18 levels were measured in SFMCs and PBMCs that were untreated, treated with TNF, or treated with monoclonal antibodies targeting TNF (anti-TNF).

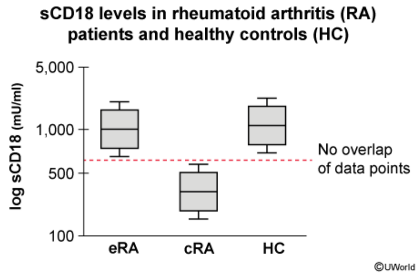
Which statement about RA is best supported by the information in the passage?

- A. eRA can be diagnosed by comparing DAS28CRP scores between patients and healthy controls. [5%]
- B. Anti-inflammatory medications have no effect on RA disease activity.
- C. cRA patients exhibit the highest LFA-1/ICAM-1 binding activity. [5%]
- ✔ D. eRA and cRA patients can be differentiated by assessing sCD18 levels. [88%]

Correct

88% answered correctly

Explanation:



A box-and-whisker plot is used to visually represent the **median** and **distribution** of the data into **quartile ranges**. It shows the following:

- The **upper quartile** (Q3; 75% of the data lies below this value), shown as the top edge of the box
- The **median** (50% of the data lies below this value), shown as the line inside the box
- The **lower quartile** (Q1; 25% of the data lies below this value), shown as the bottom edge of the box
- The **highest** and **lowest data points**, which are no more than 1.5 times the interquartile range (IQR) from the edge of the box; these points are shown as whiskers. The IQR is equal to the difference between the upper and lower quartiles (ie, the range containing the middle 50% of values).
- **Outliers**, shown as data points outside the whiskers (when such points exist in the data set)

Based on Figure 1, sCD18 levels are similar (ie, a large area of overlap between the box-and-whisker plots) in eRA patients and healthy controls but decreased in cRA patients. This result is evidenced by the lower position of the IQR in cRA

patients relative to that of eRA patients and healthy controls. In addition, no overlap is found between the IQRs of eRA and cRA groups, meaning that both groups differ in mean and data spread. Accordingly, sCD18 levels can be used to determine disease status (eRA or cRA). Therefore, Figure 1 data support the claim that **eRA** and **cRA patients** can be **differentiated** by **assessing** their **sCD18** levels.

(Choice A) The passage states that DAS28CRP scores indicate disease activity in RA, but only RA-affected individuals (*not* healthy controls) are assigned a DAS28CRP score. Therefore, comparing DAS28CRP scores between RA patients and healthy controls would *not* be possible.

(Choice B) Figure 2 shows DAS28CRP scores for eRA patients after treatment with anti-inflammatory drugs. The data demonstrate a marked reduction in DAS28CRP scores, which indicates that the anti-inflammatory drugs *reduced* (ie, had an effect on) disease activity.

(Choice C) The passage states that LFA-1/ICAM-1 binding induces CD18 release. The observed decrease in sCD18 of cRA patients implies an overall decrease in LFA-1/ICAM-1 binding. Therefore, cRA patients do *not* likely exhibit the highest LFA-1/ICAM-1 binding activity.

Educational objective:
Box-and-whisker plots are used to visually represent the median and distribution of a data set into quartile ranges.

Time spent:0 sec
Subject:
Biology

QID:400809
System:
Skin and Immune Systems

Rheumatoid arthritis (RA) is an autoimmune disease that causes chronic synovial (joint) inflammation. RA immunopathogenesis involves LFA-1, a surface receptor found on leukocytes that is composed of an α-chain (CD11a) and a β-chain (CD18). The binding of LFA-1 to ICAM-1, an adhesion protein on the surface of B cells and endothelial cells, stimulates the migration of leukocytes to specific areas of inflammation. After successful LFA-1/ICAM-1 binding, CD18 is released into the plasma and synovial fluid (SF), where it exists in soluble form (sCD18). The levels of sCD18 in SF can be used as a marker for inflammation in RA.

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Experiment 1

An immunofluorescence assay was used to measure plasma sCD18 levels in patients with eRA and cRA, and in age- and gender-matched healthy controls (HC).

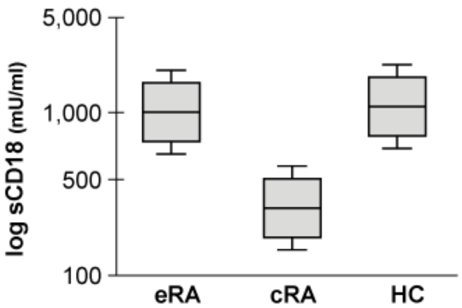


Figure 1 Baseline sCD18 measurements (Note: 50% of data points are below the median value; median = line inside the box.)

Experiment 2

Clinical assessments were performed in eRA patients to establish a baseline disease status. eRA patients were assigned a disease activity score based on characteristics of 28 joints and levels of C-reactive protein (DAS28CRP). This was followed by treatment with anti-inflammatory drugs. DAS28CRP scores were calculated at 3-month intervals for 12 months, with lower scores indicating decreased RA severity (Figure 2).

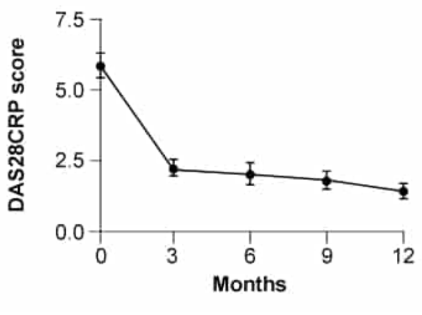


Figure 2 DAS28CRP scores in eRA patients

Experiment 3

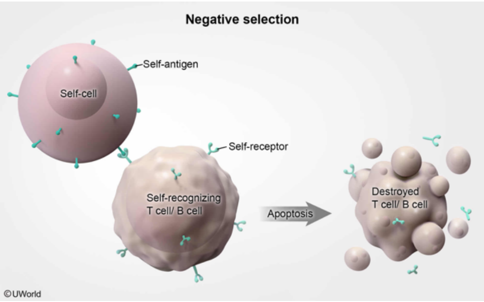
Heterogeneous populations of mononuclear cells were cultured from the synovial fluid (SFMC) and peripheral blood (PBMC) of eRA patients during an acute exacerbation of their symptoms. sCD18 levels were measured in SFMCs and PBMCs that were untreated, treated with TNF, or treated with monoclonal antibodies targeting TNF (anti-TNF).

A self-directed antibody known as rheumatoid factor (RF) has been found to be involved in RA-mediated inflammation. RF is an antibody that targets a patient's own IgG antibodies, which results in the formation of immune complexes that induce an inflammatory response. High concentrations of self-antibodies such as RF are NOT typically found in the human body because:

- A. RF functions as a complement protein. [4%]
- B. B cells and T cells that recognize self-antigens are destroyed. [75%]
- C. antigen-presenting cells cannot present self-antigens. [12%]
- D. T-cell receptor recombination cannot produce self-identifying receptors. [7%]

Correct
75% answered correctly

Explanation:



Normally, the immune system is said to be **self-tolerant** because it does not attack itself but serves to protect the body from foreign substances (antigens). During the early stages of immune cell development, which occurs in the thymus (T cells) and bone marrow (B cells), cells undergo genetic recombination to rearrange their DNA. This rearrangement leads to the expression of novel antigen-binding receptors on the surface of T cells and antibodies on the surface of B cells.

The production of billions of immune cells with varied receptor sets results in an immune system able to target many different types of antigens. However, DNA rearrangement also results in the expression of receptors that target endogenous molecules (self-antigens). To avoid rampant immune responses against the body itself, these **self-recognizing** cells are normally **destroyed**.

Negative selection is the process by which immature T cells and B cells possessing receptors that bind to self-antigens are destroyed or inactivated. These cells are either eliminated by programmed cell death (ie, apoptosis) or made unresponsive to antigens (anergic). According to the question, **rheumatoid factor** (RF) is an antibody that mediates an **autoimmune response** by targeting a patient's own IgG antibodies. **B cells and T cells that mediate recognition** and destruction of **self-antigens** are typically **destroyed** during negative selection. This means that antibodies such as RF directed at self-antigens are *not* typically found in the human body.

(Choice A) Complement proteins are blood proteins that increase the effectiveness of antibodies by helping recruit and activate phagocytes. The question describes RF as an antibody that targets self-IgG antibodies; therefore, RF is *not* a complement protein.

(Choices C and D) Rearrangement of DNA leads to the expression of surface antibodies/receptors that *can identify* self-antigens on B cells and T cells. Identification of self-antigens by B cells enables subsequent *presentation* of the self-antigens. However, these self-recognizing cells are normally *destroyed* during immune cell maturation (ie, during negative selection).

Educational objective:
Autoimmunity disorders occur due to the immune system's failure to identify and destroy immune cells that recognize self-antigens. Self-reactive B cells and T cells are normally eliminated in the bone marrow and thymus, respectively.

Time spent:0 sec	**QID**:400810
Subject:	**System:**
Biology	Skin and Immune Systems

Rheumatoid arthritis (RA) is an autoimmune disease that causes chronic synovial (joint) inflammation. RA immunopathogenesis involves LFA-1, a surface receptor found on leukocytes that is composed of an α-chain (CD11a) and a β-chain (CD18). The binding of LFA-1 to ICAM-1, an adhesion protein on the surface of B cells and endothelial cells, stimulates the migration of leukocytes to specific areas of inflammation. After successful LFA-1/ICAM-1 binding, CD18 is released into the plasma and synovial fluid (SF), where it exists in soluble form (sCD18). The levels of sCD18 in SF can be used as a marker for inflammation in RA.

CD18 release has been shown to increase in response to tumor necrosis factor (TNF), a pro-inflammatory cytokine involved in the inflammatory response of RA. In patients with newly diagnosed and untreated RA (eRA), leukocytes produce adequate quantities of LFA-1 and plasma sCD18 levels are unchanged (ie, elevated). However, patients with chronic RA (cRA) experience dysregulation of LFA-1 synthesis and an increase in competition for ICAM-1 by other ligands. Over time, this leads to a reduction in CD18 shedding and, as a result, a decrease in sCD18.

Experiment 1

An immunofluorescence assay was used to measure plasma sCD18 levels in patients with eRA and cRA, and in age- and gender-matched healthy controls (HC).

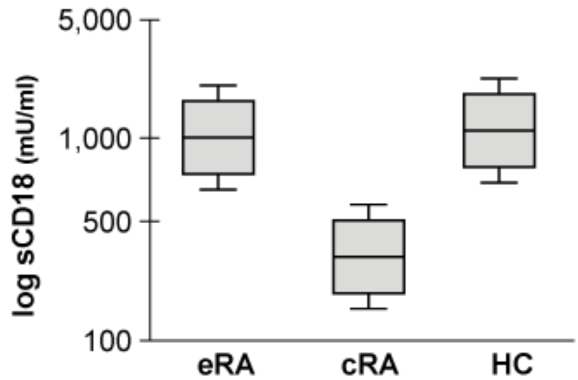


Figure 1 Baseline sCD18 measurements (Note: 50% of data points are below the median value; median = line inside the box.)

Experiment 2

Clinical assessments were performed in eRA patients to establish a baseline disease status. eRA patients were assigned a disease activity score based on characteristics of 28 joints and levels of C-reactive protein (DAS28CRP). This was followed by treatment with anti-inflammatory drugs. DAS28CRP scores were calculated at 3-month intervals for 12 months, with lower scores indicating decreased RA severity (Figure 2).

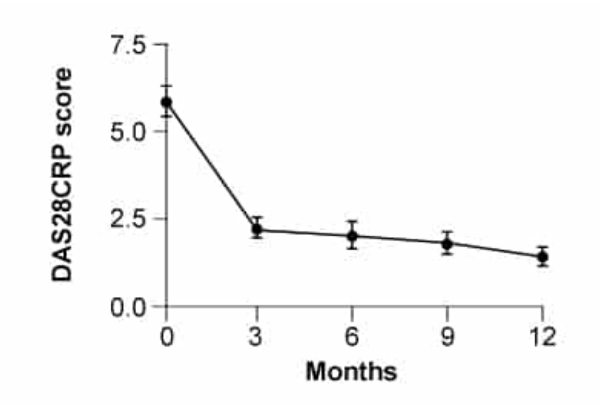


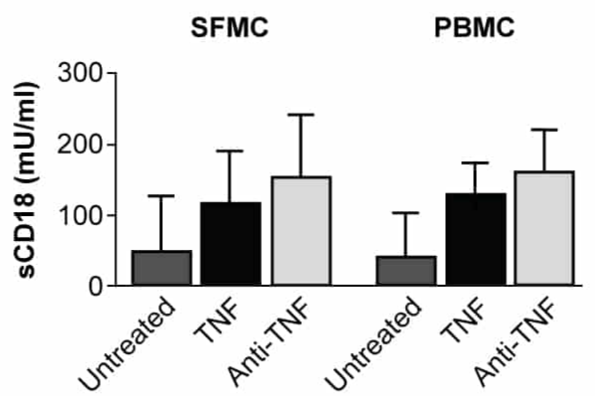
Figure 2 DAS28CRP scores in eRA patients

Experiment 3

Heterogeneous populations of mononuclear cells were cultured from the synovial fluid (SFMC) and peripheral blood (PBMC) of eRA patients during an acute exacerbation of their symptoms. sCD18 levels were measured in SFMCs and PBMCs that were untreated, treated with TNF, or treated with monoclonal antibodies targeting TNF (anti-TNF).

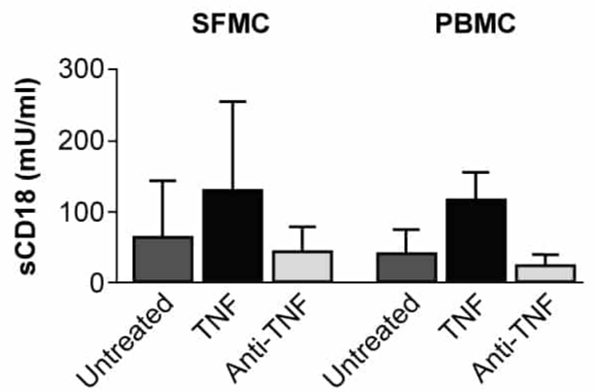
Which graphic shows the expected results of Experiment 3?

A.



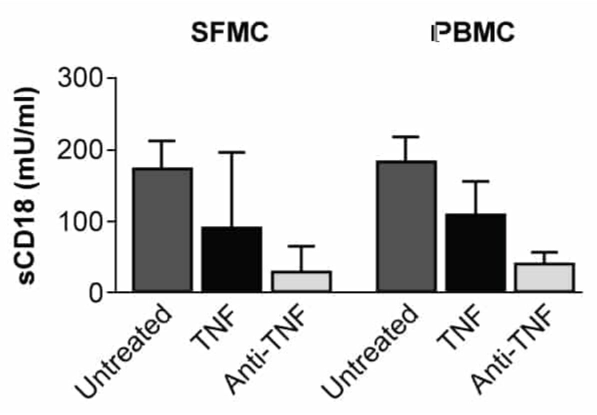
[4%]

B.



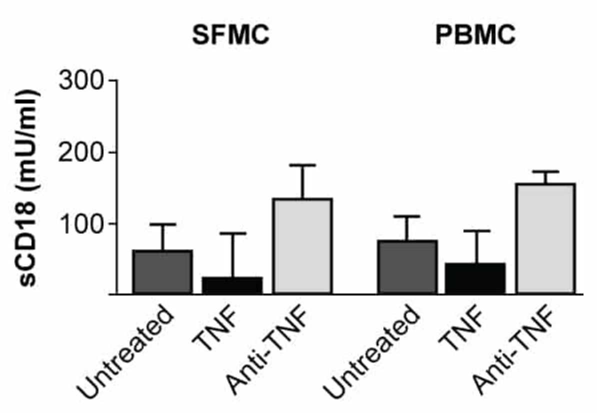
[69%]

C.



[17%]

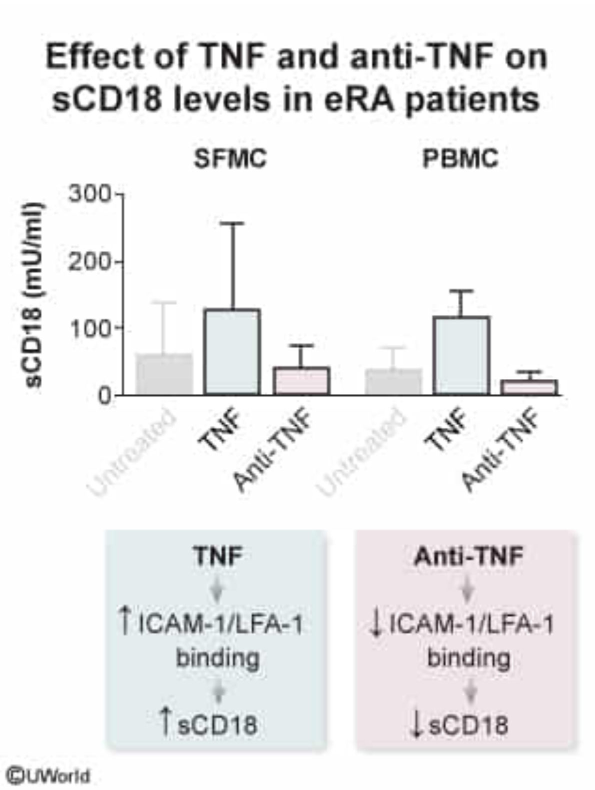
◯ D.



[7%]

Correct
70% answered correctly

Explanation:



In Experiment 3, heterogeneous populations of mononuclear cells from the synovial fluid (SFMC) and peripheral blood (PBMC) of eRA patients were cultured to measure sCD18 levels in response to no treatment, treatment with TNF, or treatment with monoclonal antibodies targeting TNF (ie, anti-TNF). According to the passage, **TNF**

stimulates the **release of CD18** from the cell surface, promoting an increase in its soluble form, sCD18. Consequently, **TNF-treated mononuclear cells** would be expected to have significantly **increased sCD18 concentrations** compared to the untreated group.

The passage describes treating cultured cells with a **monoclonal antibody targeting TNF** (anti-TNF), which likely prevents TNF from interacting with the cultured cells. Therefore, the **cells treated with anti-TNF** would be expected to have **decreased levels of sCD18** because the antibody treatment inhibits the effects of endogenous TNF (ie, increased sCD18 levels).

In summary, the expected result of Experiment 3 would show the **highest sCD18 levels** in the **TNF-treated cells**, the **lowest levels** in the **anti-TNF-treated cells**, and the **untreated cells** with **sCD18 levels between the treated cell types** because of endogenous TNF action. This trend would be expected to hold in both SFMCs and PBMCs.

(Choice A) This graph shows the anti-TNF–treated cells having a higher sCD18 level than both the untreated and the TNF-treated cells. Treatment with anti-TNF would be expected to *decrease* sCD18 levels below the level of other groups due to the ability of the antibody to sequester TNF, which inhibits its molecular action (ie, stimulating CD18 release).

(Choice C and D) TNF-treated mononuclear cells would be expected to have an increased (*not* decreased) concentration of sCD18 compared to the untreated group because of the ability of TNF to induce the release of CD18.

Educational objective:
Monoclonal antibodies can sequester specific molecules and subsequently inhibit their molecular action. Monoclonal antibodies are derived from a single clone, and all recognize a single epitope.

Time spent:0 sec	**QID**:400811
Subject:	**System:**
Biology	Skin and Immune Systems